***Situation:***

Let us look at the family tree of a male drone bee and a female bee as shown Fig 7.19.

Here, female bees have 2 parents, male (drone) bees have just one parent, a female. (Male bee (drone) are produced by the queen's unfertilized eggs, so male(drone) bees only have a mother but no father!)

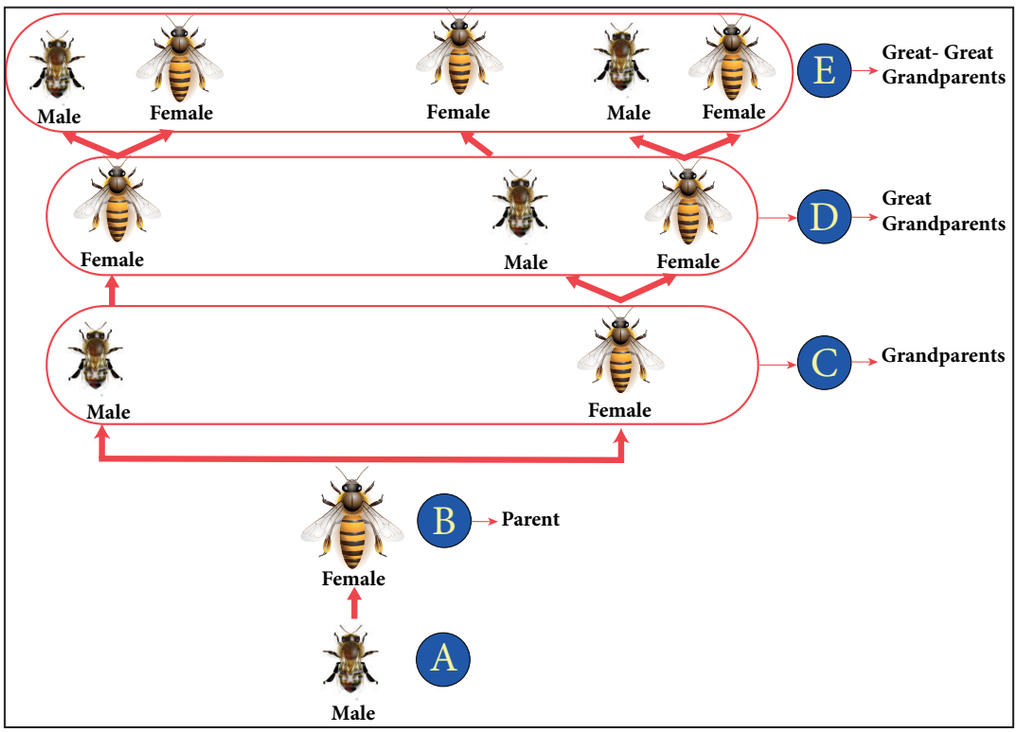



Fig. 7.19

From the picture the following points are noted:

1. The male Ⓐ has 1 parent, a female Ⓑ.
2. The male Ⓐ has **2** grandparents Ⓒ, since his mother had parents Ⓒ, a male and a female.
3. The male Ⓐ has **3** great-grandparents Ⓓ: since his grandmother has two parents but his grandfather has only one.

Now, answer, how many great-great-grandparents did the male Ⓐ have?

Let us try to find the relationship among the pattern of bees family by representing in the tabular form given below,

| Number of Ⓐ | Parents Ⓑ | Grandparents Ⓒ | Great Grandparents Ⓓ | Great- Great Grandparents Ⓔ | Great- Great-Great Grandparents Ⓕ |
|---|---|---|---|---|---|
| a Male bee (1) | 1 | 2 | 3 | 5 | 8 |
| a Female bee (1) | 2 | 3 | 5 | 8 | 13 |

We see the Fibonacci numbers 1, 1, 2, 3, 5, 8, 13… in the above table.



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